

Financial Analysis Report

Part A - Project overview

Project Name: Development of a Mobile BaYenking Application

Objective: Develop a mobile banking application to enhance customer convenience and reduce operational costs.

Expected ROI: 25% within two years.

Initial Investment: \$500,000

Project Duration: 12 months

Estimated Revenue: \$200,000 per year

Estimated Savings: \$50,000 per year in operational costs

Part B - Project baseline analysis

Original Baselines

Scope Baseline:

- Fully functional mobile banking app with core banking features.
- Integration with existing banking systems.

Schedule Baseline:

- Planning Phase: 1 month
- Development Phase: 6 months
- Testing Phase: 3 months
- Deployment Phase: 2 months

Cost Baseline:

- Planning: \$50,000
- Development: \$300,000
- Testing: \$100,000
- Deployment: \$50,000

Updated Baselines

After 6 months, additional security features required by new regulations extended the development duration and increased costs.

Updated Scope Baseline:

- Fully functional mobile banking app with core banking features.
- Include Additional Security Features.
- Integration with existing banking systems.

Updated Schedule Baseline:

- Planning Phase: 1 month
- Development Phase: 8 months
- Testing Phase: 3 months
- Deployment Phase: 2 months
- Total Project Duration: 14 months

Updated Cost Baseline:

- Planning: \$50,000
- Development: \$400,000
- Testing: \$100,000
- Deployment: \$50,000
- Total Project Cost: \$600,000

Part C - Project ETC and EAC calculations

Planning: \$50,000

Development: \$300,000

Testing: \$100,000

Deployment: \$50,000

After 6 months into the project, the team realized that additional security features were needed due to new regulatory requirements. Based on the team's input, you arrived at a revised estimate that the development phase will be extended by 2 months, and the budget will be increased by \$100,000.

Estimate to Complete (ETC): \$250,000

The remaining work in the project includes the new scope as well. Therefore, the new scope cost of \$100,000 is also considered for calculating Estimate to Complete (ETC).

Estimate to Complete (ETC) for remaining work = New scope Development Cost + Testing Cost + Development Cost

$$\text{ETC} = \$100,000 + \$100,000 + \$50,000$$

Estimate at Completion (EAC): \$600,000

Formula: $\text{EAC} = \text{AC} + \text{ETC}$

Assuming that the project is going as per plan, Actual Cost (AC) spent by the end of 6 months is: Planning Cost + Original Development Cost

$$\text{AC} = \$50,000 + \$300,000$$

Actual Cost = \$350,000

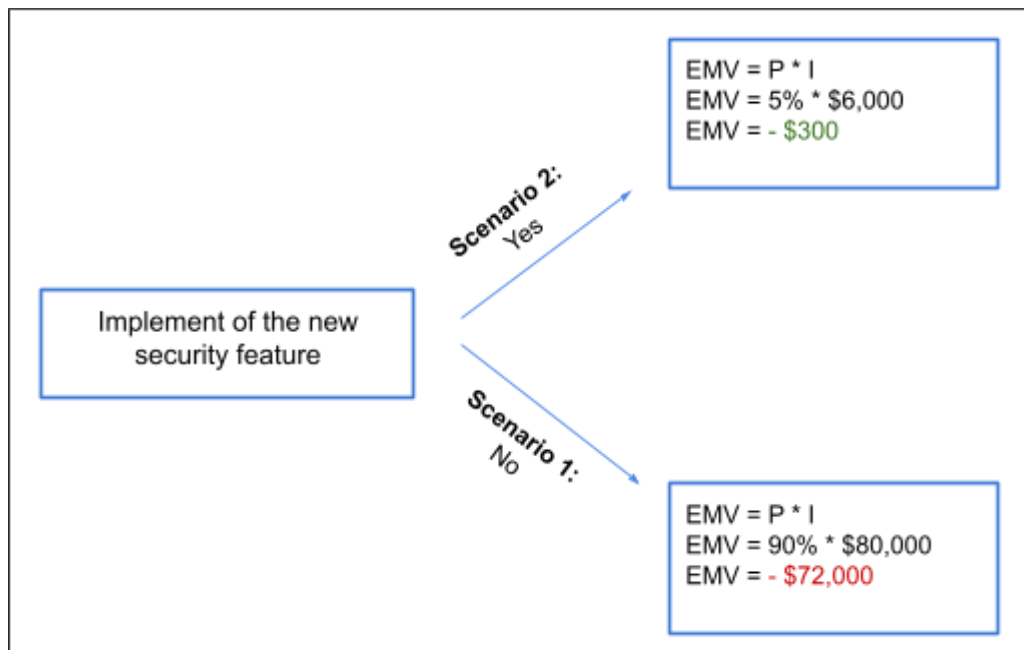
Applying the EAC Formula,

$$EAC = AC + ETC = \$350,000 + 250,000 = \$600,000$$

Part D - EMV Calculations

After 6 months into the project, the team realized that additional security features were needed due to new regulatory requirements. Based on the team's input, you arrived at a revised estimate that the development phase will be extended by 2 months, and the budget will be increased by \$100,000.

To know the absolute necessity of this extra scope, you seek experts who warn that if this feature is not included in the mobile application, it will invite a severe threat, with a probability of 90%. If it occurs, it may incur a \$80,000 impact on the business, whereas investing in the new feature would mitigate the risk to a 5% occurrence and an impact of \$6,000 when occurred.



Decision-making scenario 1: The impact of Risk occurrence when the new feature is decided not to be included in the software.

Formula: $EMV = \text{Probability (P) of Occurrence} * \text{Impact (I) If Occured}$

$$EMV = P * I = 90\% * \$80,000$$

EMV = - \$72,000 (-ve because this is a threat / impact we are measuring)

As it is the threat that we are measuring, EMV takes a -ve sign to indicate loss when the risk is occurred.

Decision-making scenario 2: The impact of Risk occurrence when the new feature is decided to be included in the software.

Formula: $EMV = \text{Probability (P) of Occurrence} * \text{Impact (I) If Occured}$

$$EMV = P * I = 5\% * \$6,000$$

EMV = - \$300 (-ve because this is a threat / impact we are missing measuring)

EMV results state that the impact of not adding the new scope (scenario 1) is huge when compared to the scenario of adding the feature. Therefore, it is best to decide implementing the new security feature and it is advisable to invest extra time and cost to accommodate the new feature.

Part E - PERT / Three-Point Estimation

Formula: Expected Time (T) = (O + 4M + P) / 6

Optimistic time (O): 7 months

Most likely time (M): 8 months

Pessimistic time (P): 9 months

$$T = (7 + (4 * 8) + 9) / 6 = 48 / 6 = 8 \text{ months}$$

The updated schedule of 8 months of developing a new scope with additional security features is accurate based on PERT estimation.

Part F - Project ROI Calculation

Expected ROI: 25% within two years.

Initial investment: \$500,000

Project duration: 12 months

Estimated revenue: \$200,000 per year

Estimated savings: \$50,000 per year in operational costs

Total Revenue and Savings over 2 years:

$$\text{Total revenue} = 2 * \$200,000 = \$400,000$$

$$\text{Total savings} = 2 * \$50,000 = \$100,000$$

$$\text{Total benefits} = \$400,000 + \$100,000 = \$500,000$$

Net Profit: - \$100,000

Formula: Net Profit = Total Benefits - Total Project Cost

$$\text{Net Profit} = \$500,000 - \$600,000 = - \$100,000$$

ROI: - 20%

Formula: $ROI = (Net\ Profit / Initial\ Investment) * 100$

$ROI = (- \$100,000 / \$500,000) * 100 = -20\%$

Part G - Project Cost - Benefit Ratio (CBR) Analysis

Cost - Benefit Ratio (CBR): 0.83

Formula: $CBR = Total\ Benefits / Total\ Project\ Cost$

$CBR = \$500,000 / \$600,000 = 0.83$

Part H - Inference

Based on the updated baselines and the ROI calculation:

- The project will not meet the initial ROI of 25%.
- The ROI is projected to be -20%, indicating a loss.
- The Cost - Benefit Ratio is less than 1 (0.83), suggesting that the benefits do not outweigh the costs.

Part I - Recommendations

Given the negative ROI and the Cost - Benefit Ratio below 1, investing in similar projects in the future is not advisable unless additional measures are taken to control costs or increase revenues. Careful consideration of potential risks and extra costs is essential in project planning and execution to ensure a positive return on investment.